

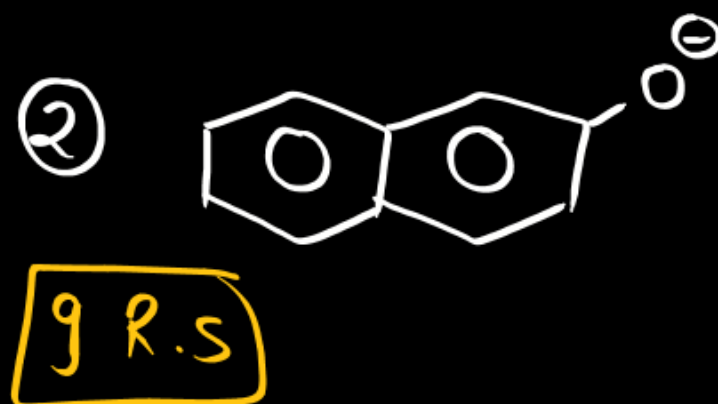
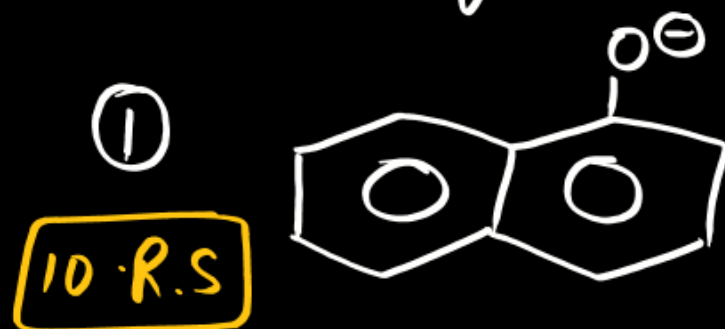
CATS

COMPETISHUN ADVANCED
TRAINING SESSION



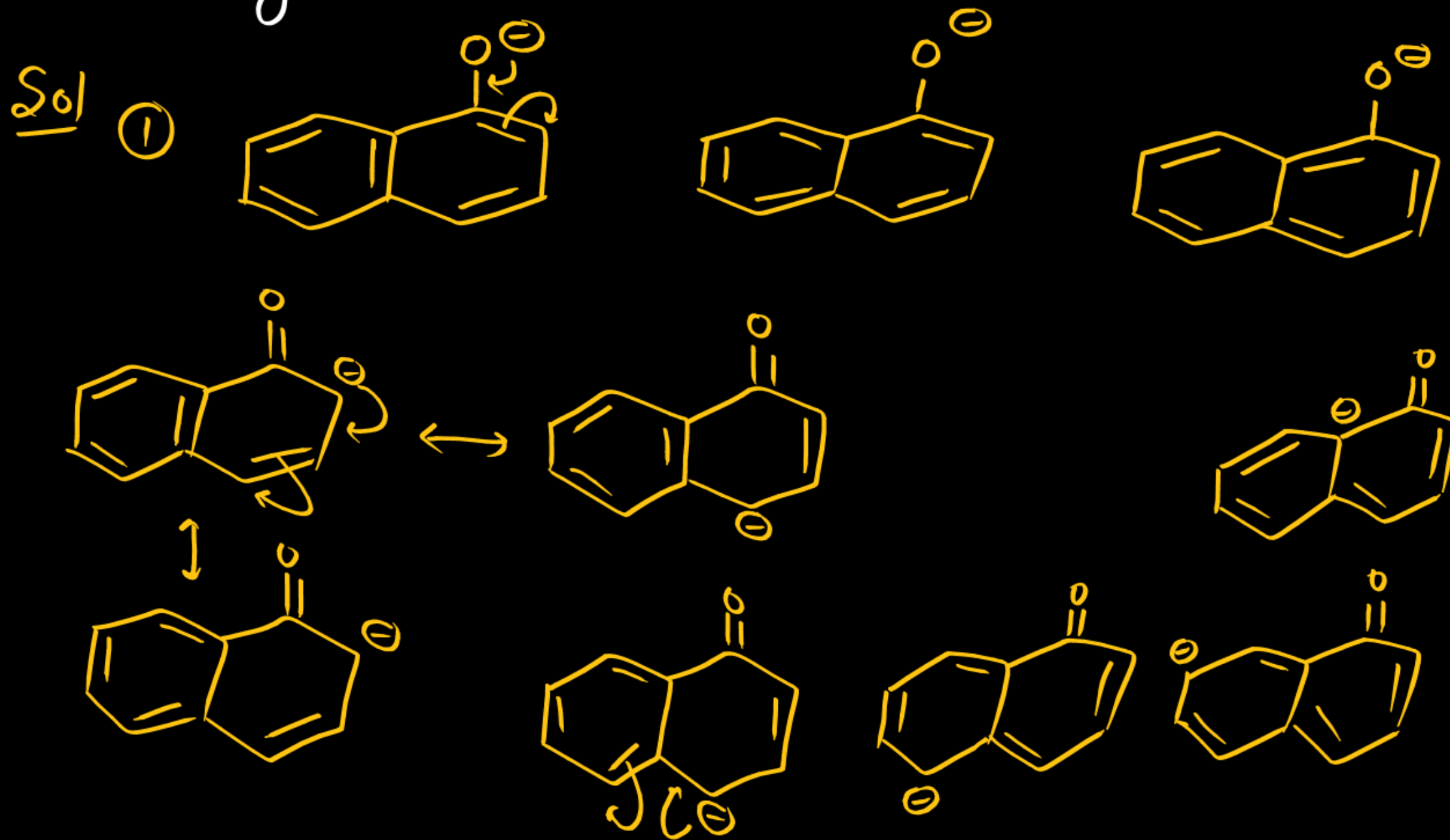
CATS (COMPETISHUN ADVANCED TRAINING SESSION)

Q. Identify no. of resonating structures :-



Stability ① > ②

acidic strength $\Rightarrow \alpha\text{-Naphthal} > \beta\text{-Naphthal}$

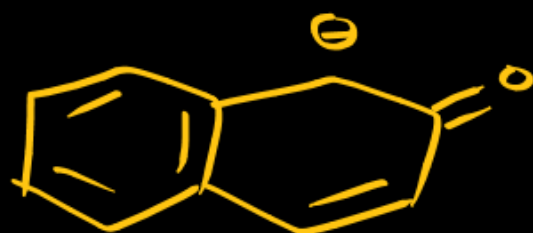
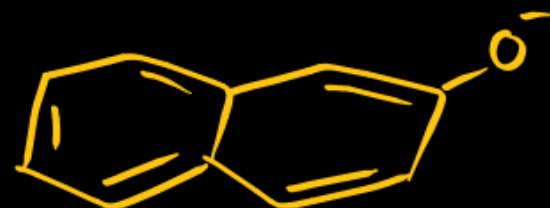


CATS (COMPETISHUN ADVANCED TRAINING SESSION)

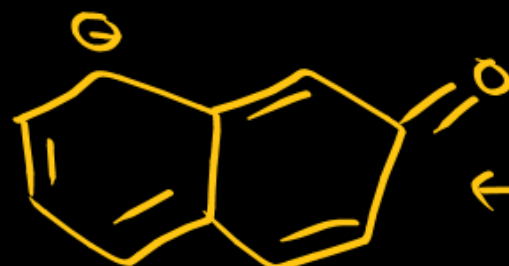
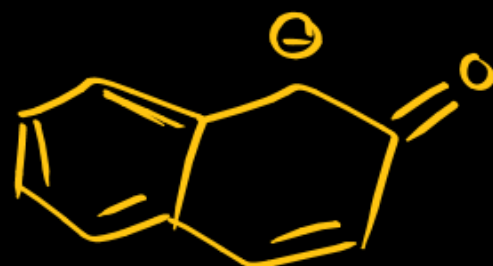
Sol/ ②



↓



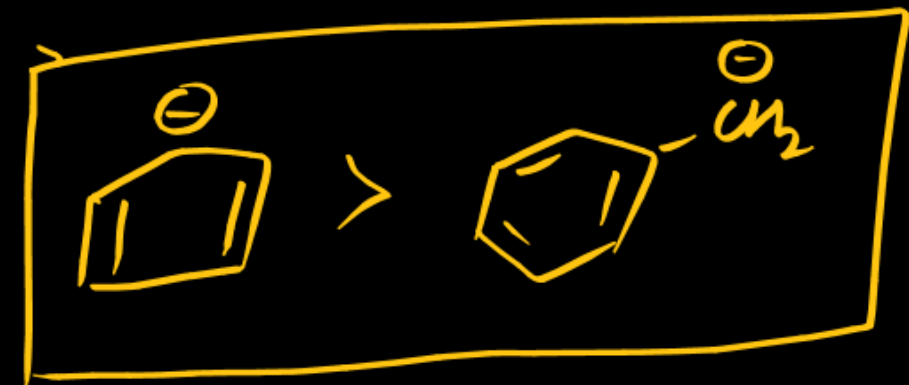
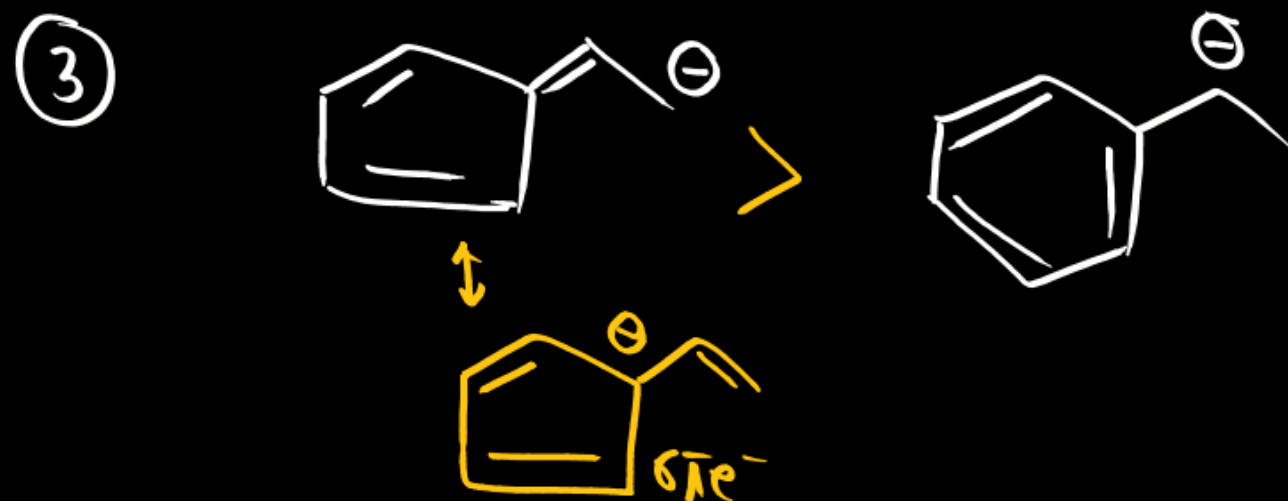
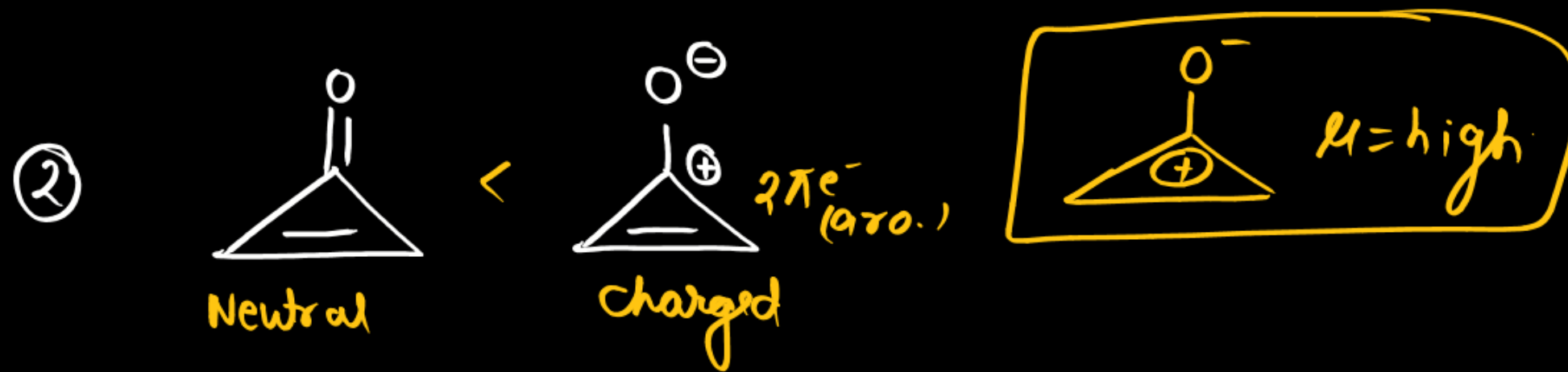
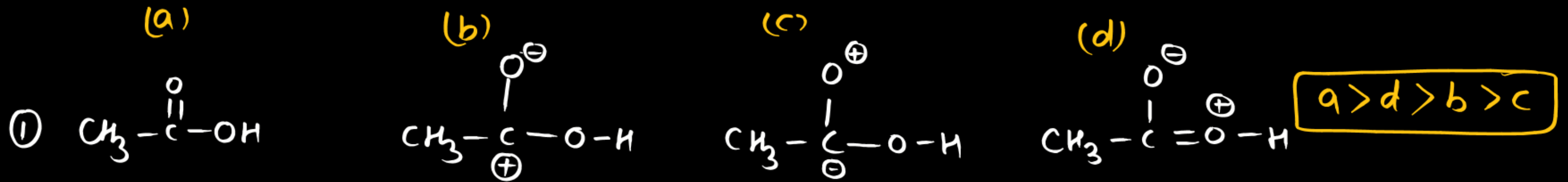
↓



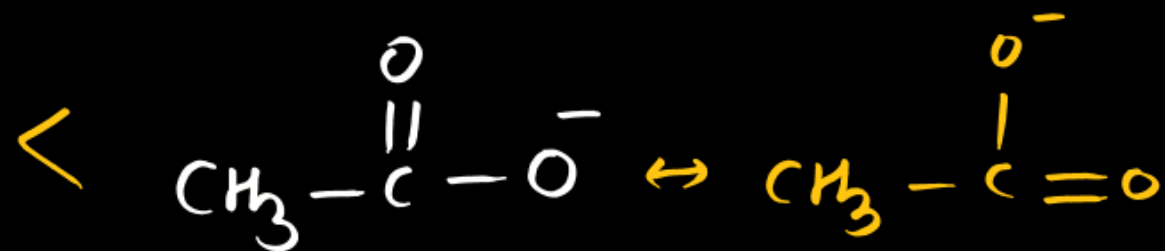
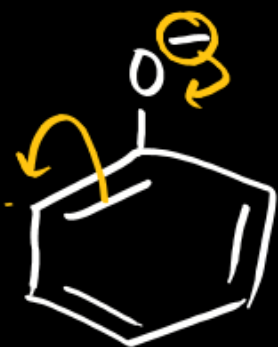
9 R.S

CATS (COMPETISHUN ADVANCED TRAINING SESSION)

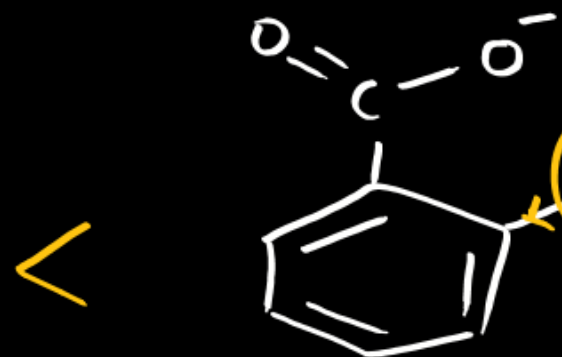
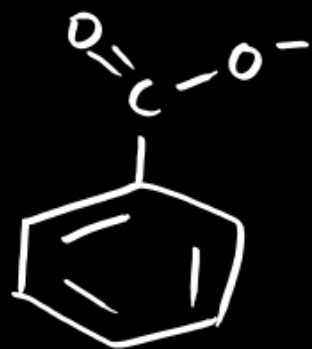
Q. Identify order of stability:-



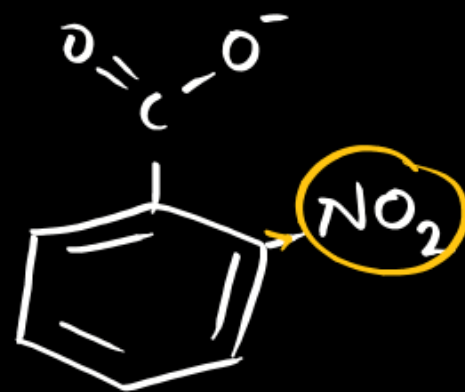
(4)



(5)

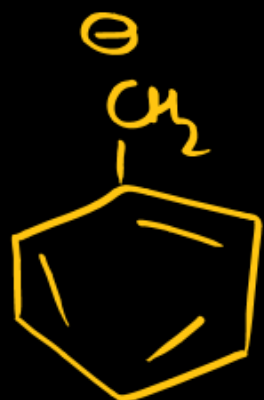


ortho effect



ortho effect

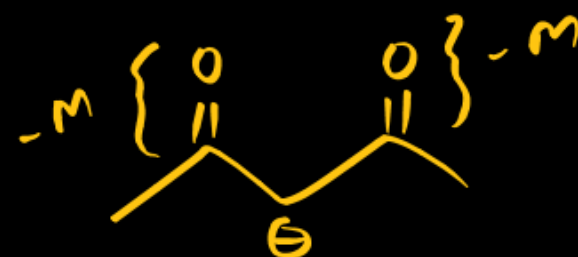
$\Rightarrow$



$<$



$<$



CATS (COMPETISHUN ADVANCED TRAINING SESSION)

Q. Give Order of heat of hydrogenation:-

(a)

(b)

(c)

Heat of hyd. $\propto$ no. π bonds
 $\propto \frac{1}{\text{Stability of alkene}}$ (same π)

(1)



(d)



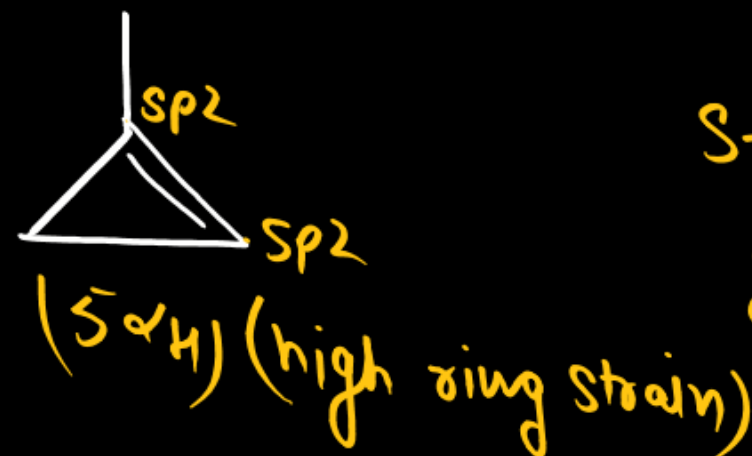
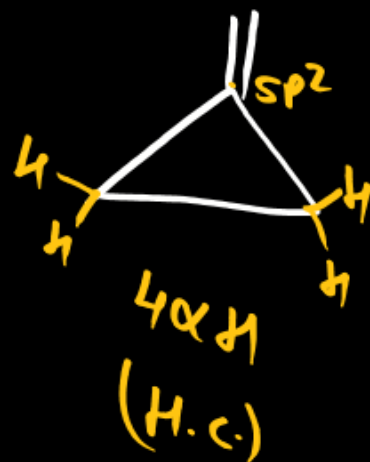
Stability = $b > a > d > c$
 H.O.H = $b < a < d < c$

(2)



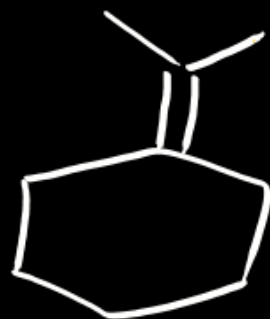
$\Rightarrow$ Stability = $a > c > b$
 $\Rightarrow$ H.O.H $\Rightarrow a < c < b$

(3)



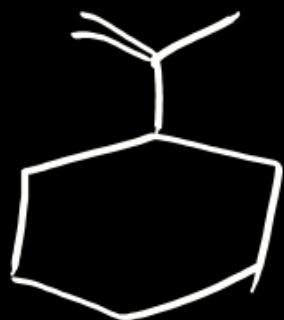
Stability $a > b$
 H.O.H = $a < b$

(4)



10 α H

H.C



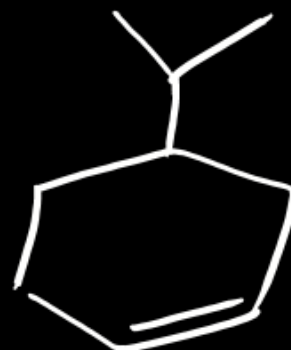
4 α H

for H.C.



5 α H

for H.C



4 α H

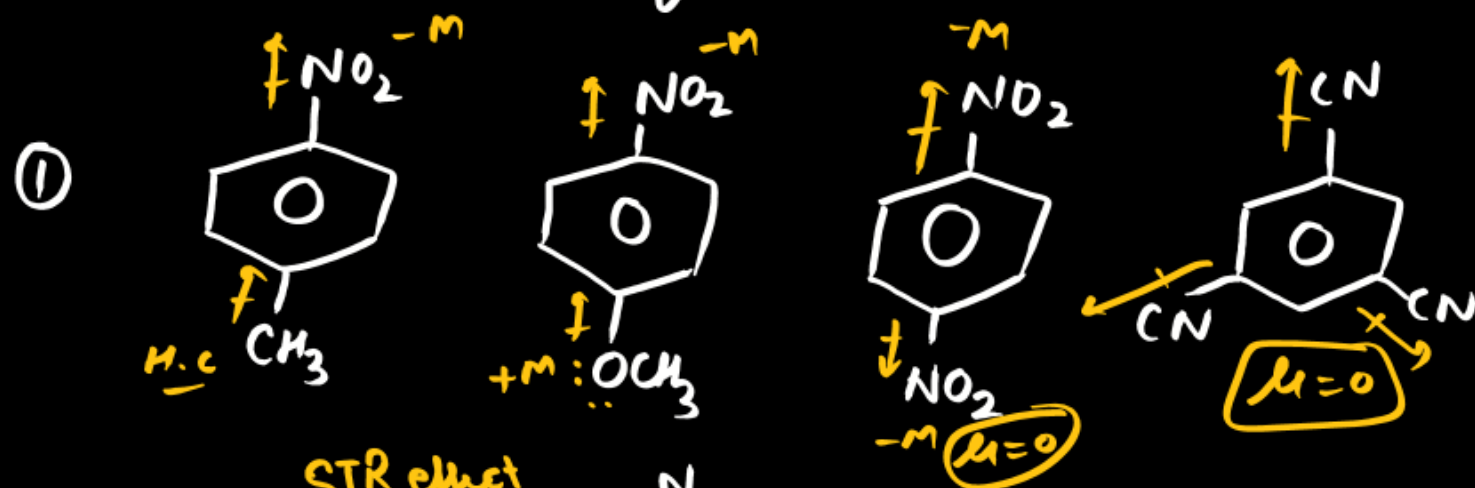
H.C.

Stability = $a > c > \underline{d} > b$

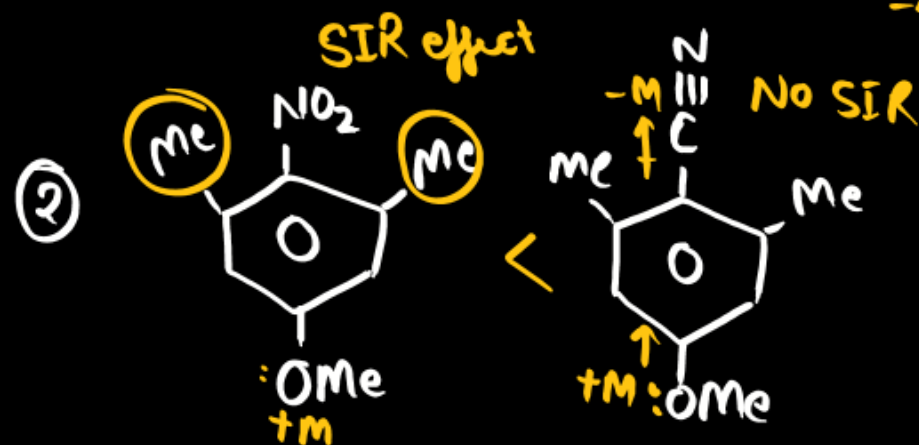
H O H = $a < c < d < b$

CATS (COMPETISHUN ADVANCED TRAINING SESSION)

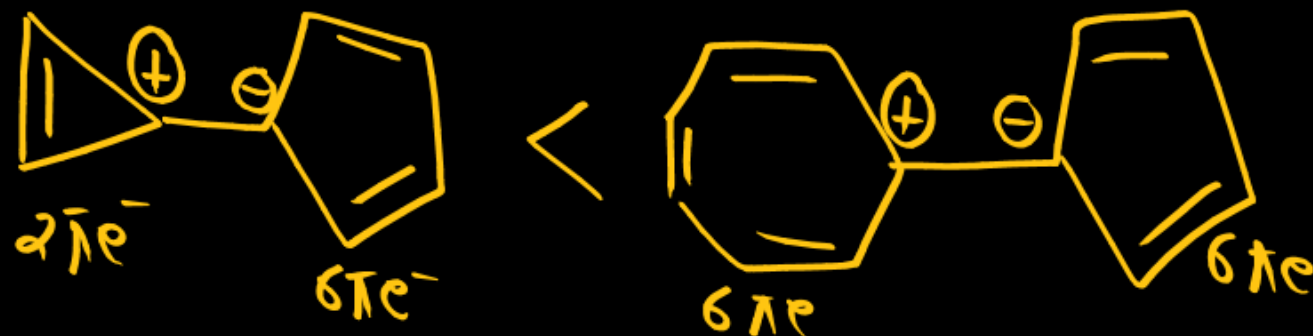
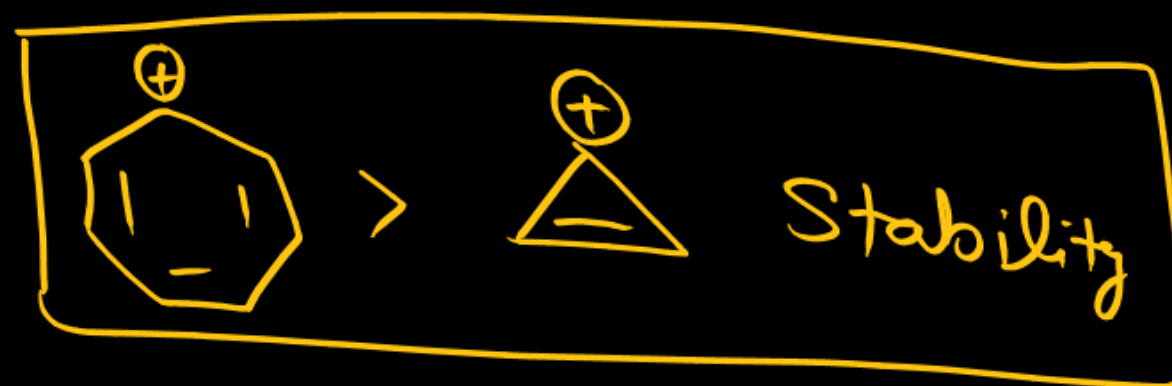
Q. Give order of dipole moment.

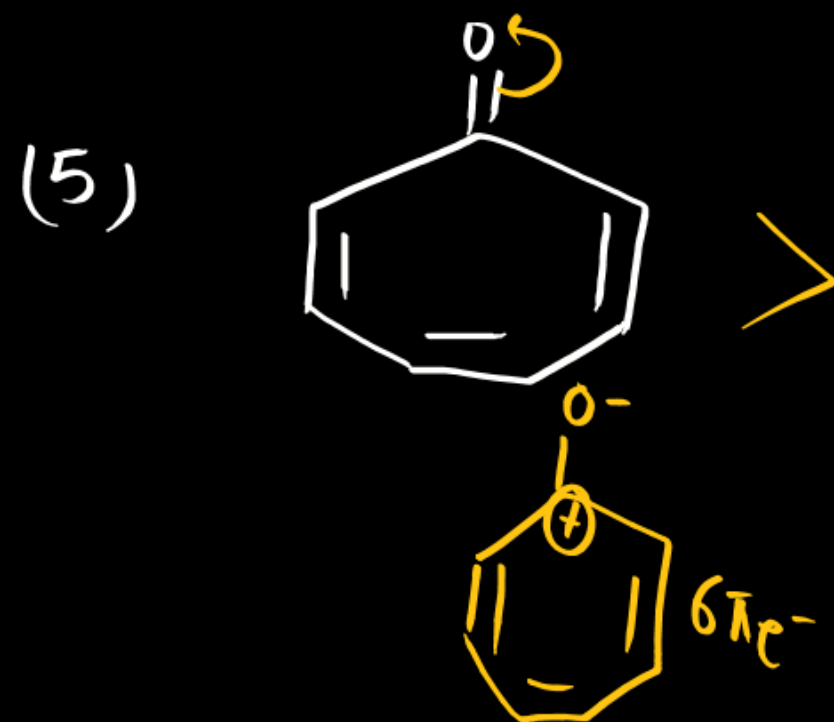
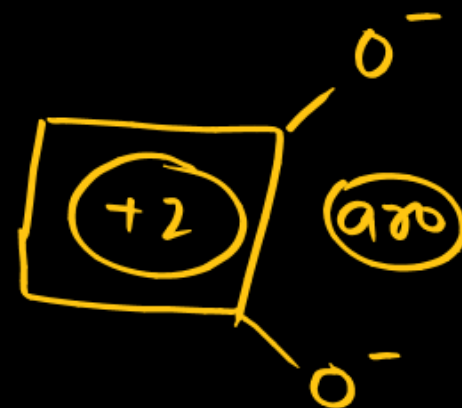
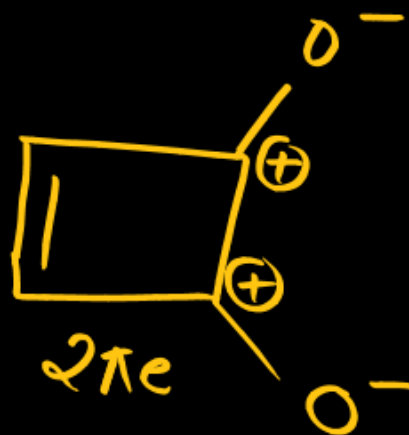
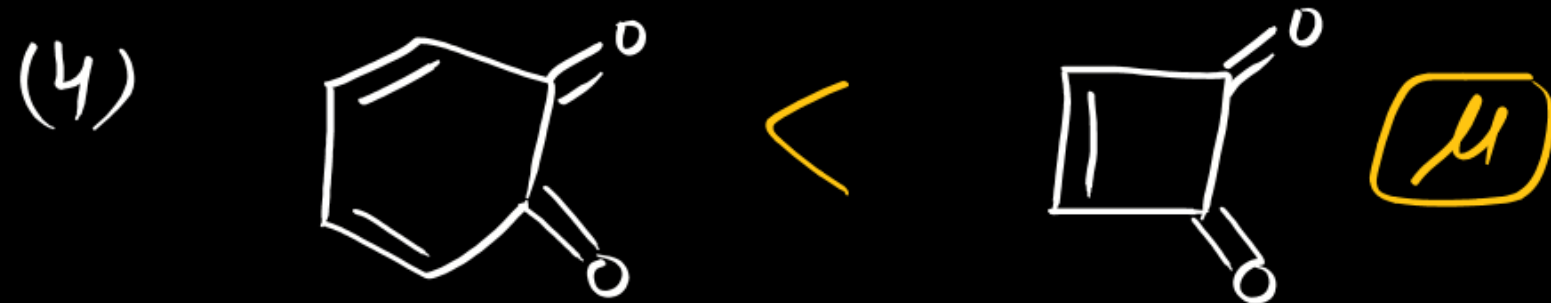


$$b > a > c = d$$



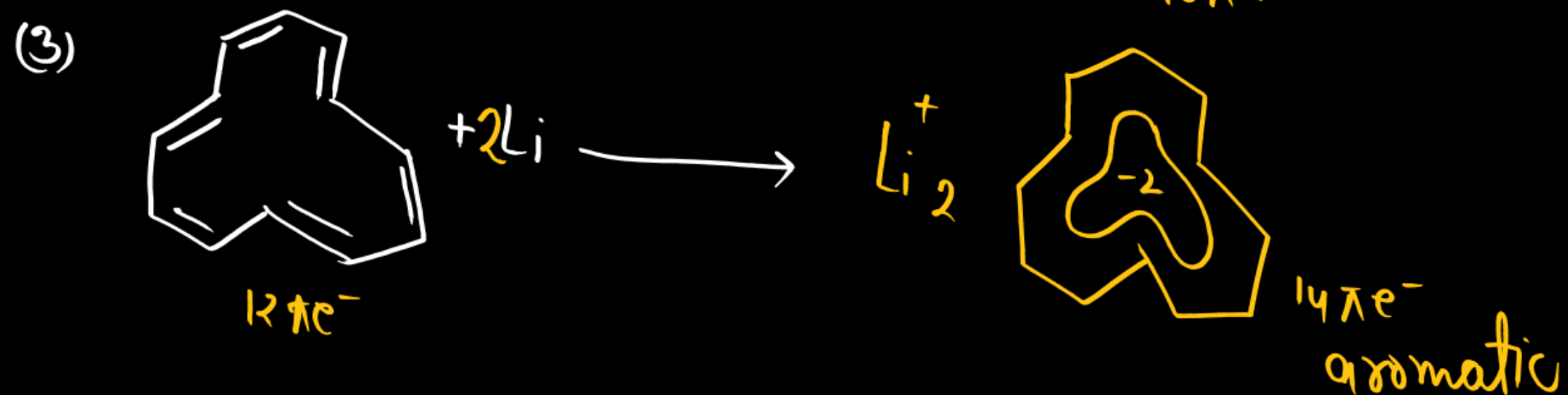
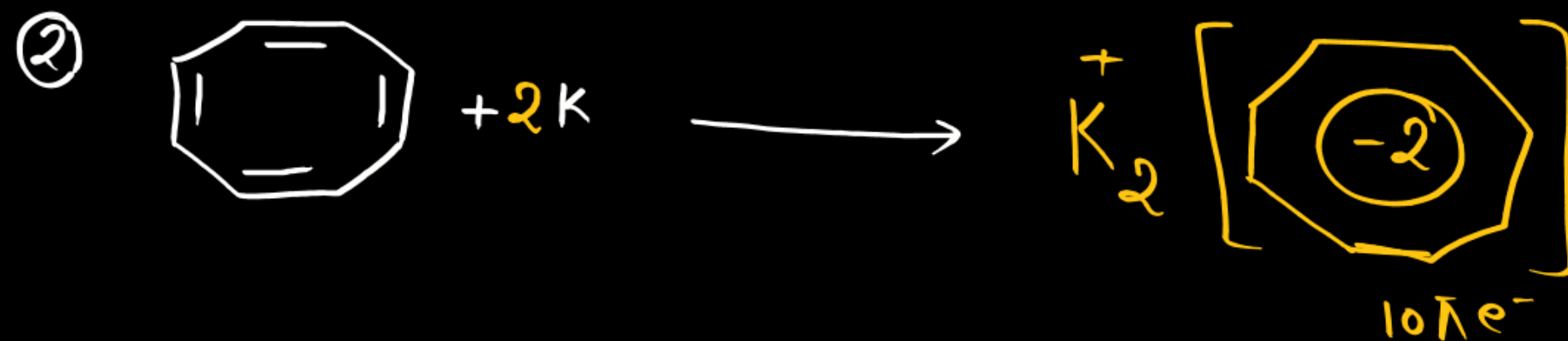
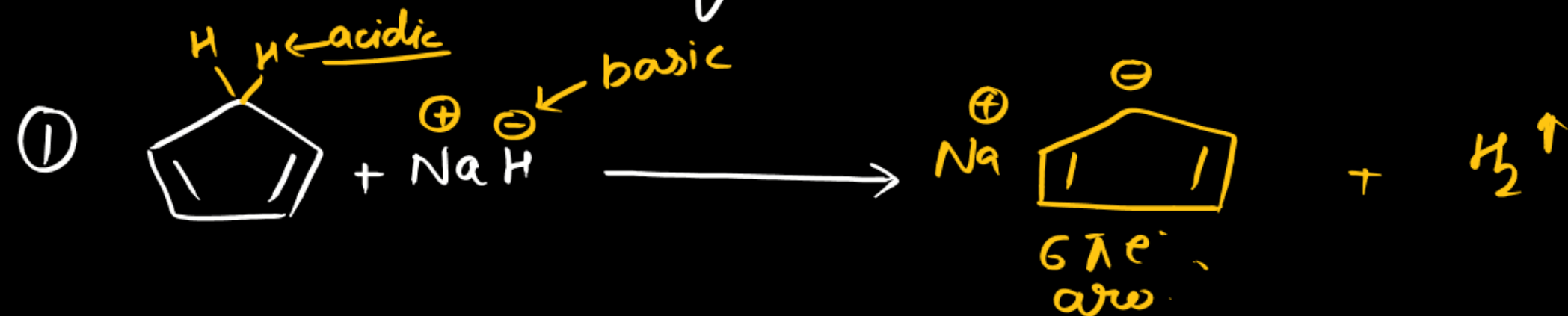
$$-M \mid -\text{NO}_2 > \text{CN}$$





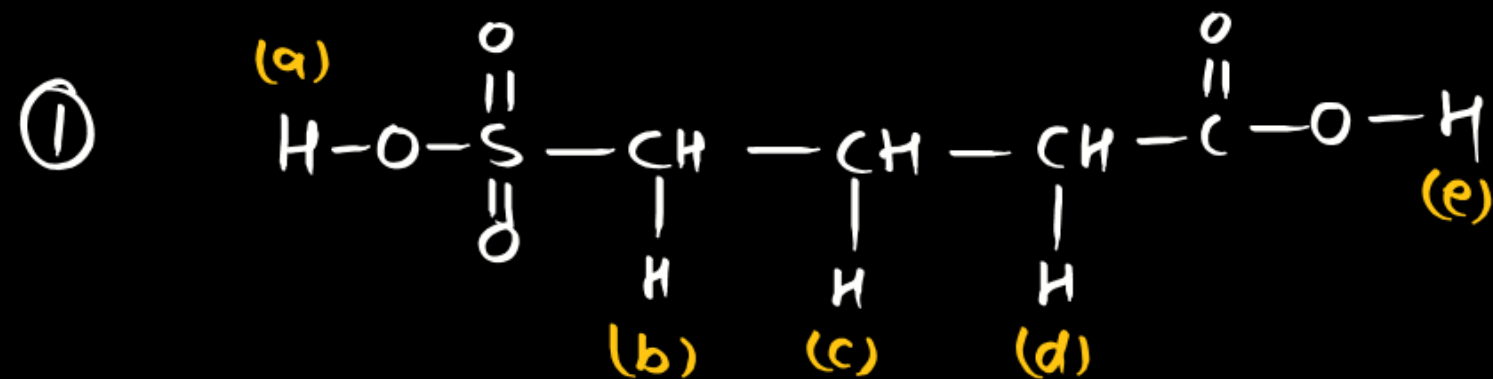
CATS (COMPETISHUN ADVANCED TRAINING SESSION)

Q. Complete the following reaction.

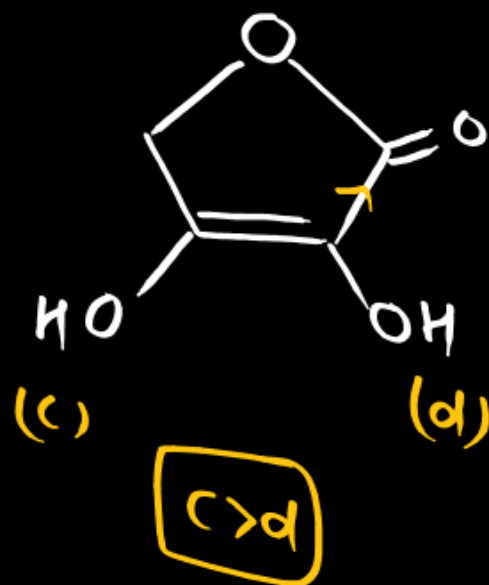
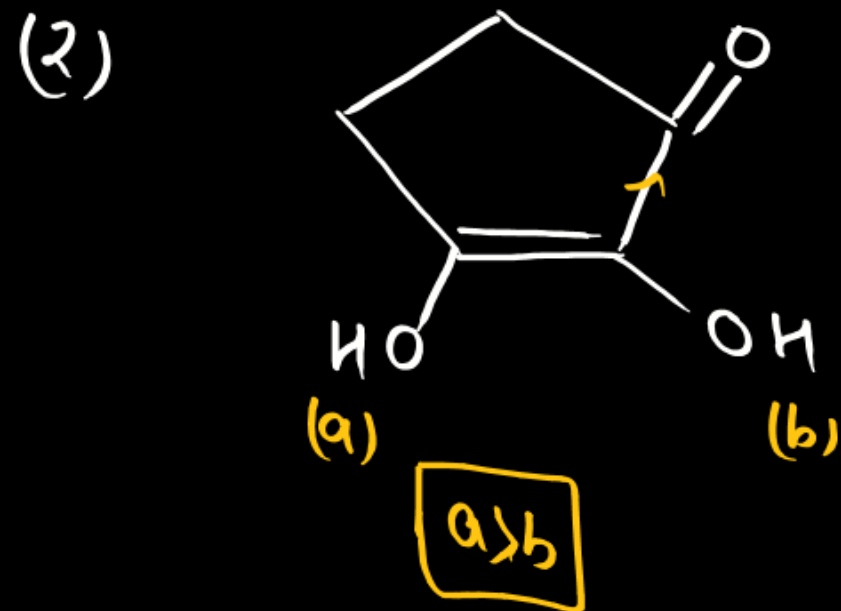
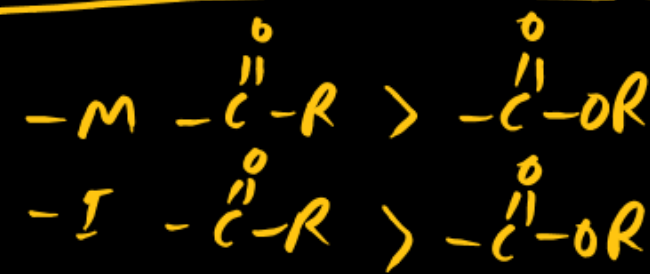


CATS (COMPETISHUN ADVANCED TRAINING SESSION)

Q. Give order of acidic strength.



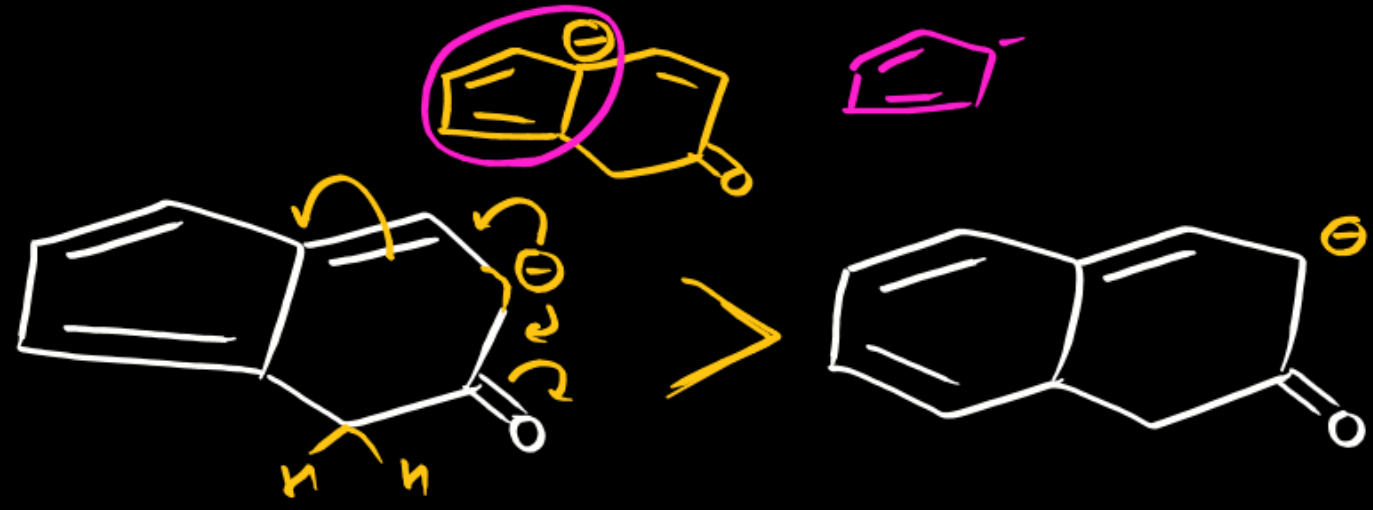
acidic $\Rightarrow a > e > b > d > c$
strength



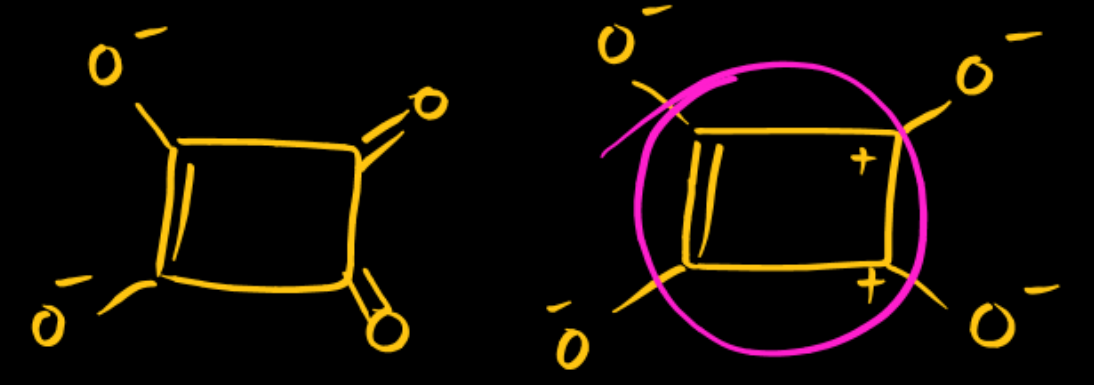
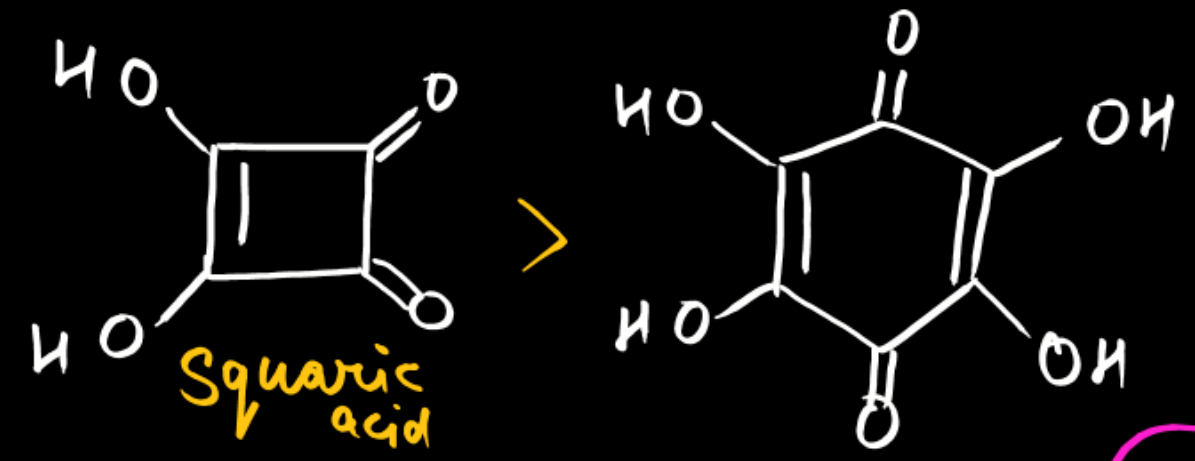
acidic = $a > c > b > d$

CATS (COMPETISHUN ADVANCED TRAINING SESSION)

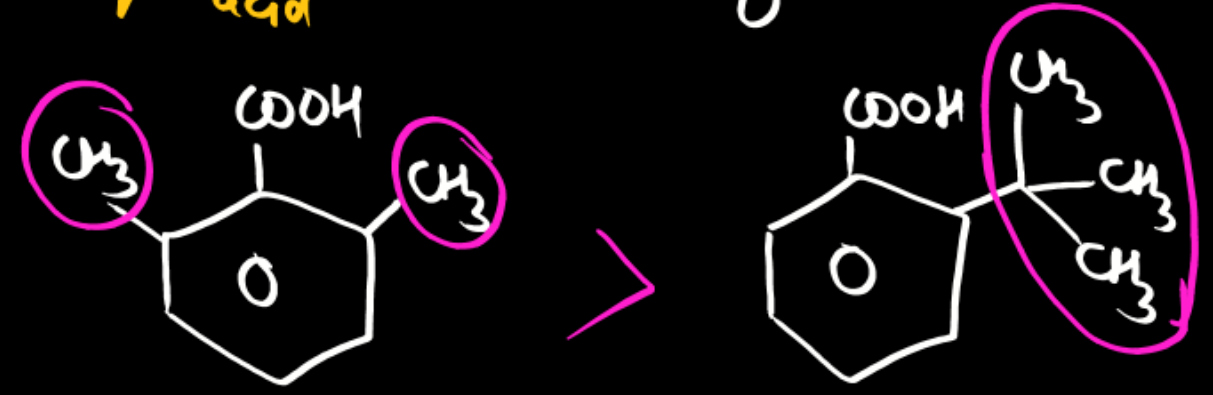
(3)



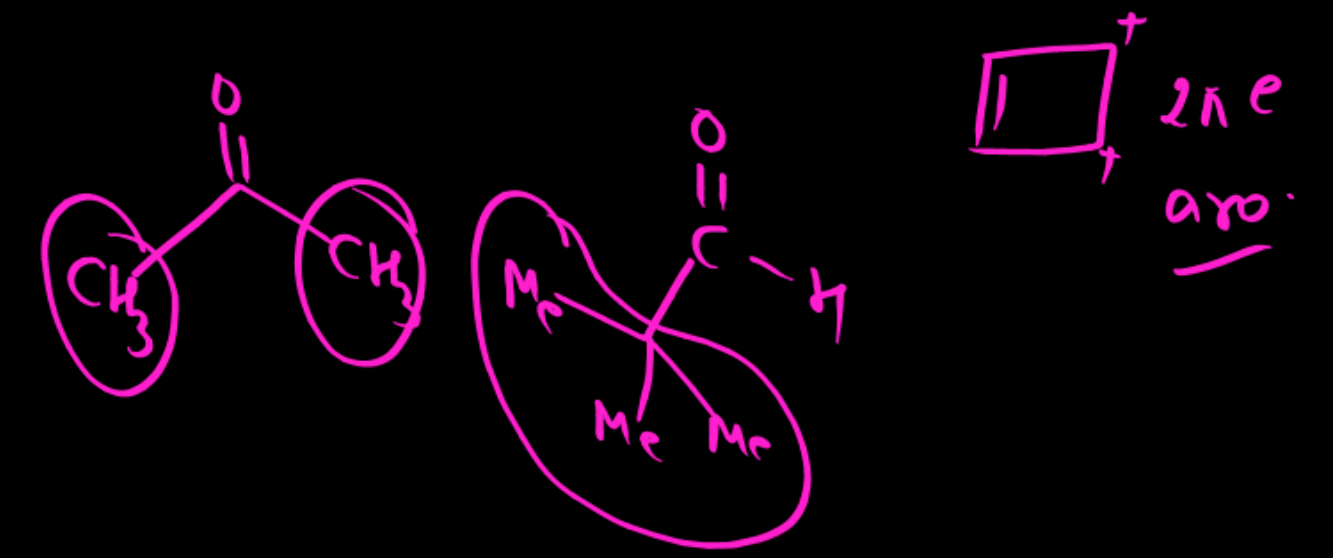
(4)



(5)



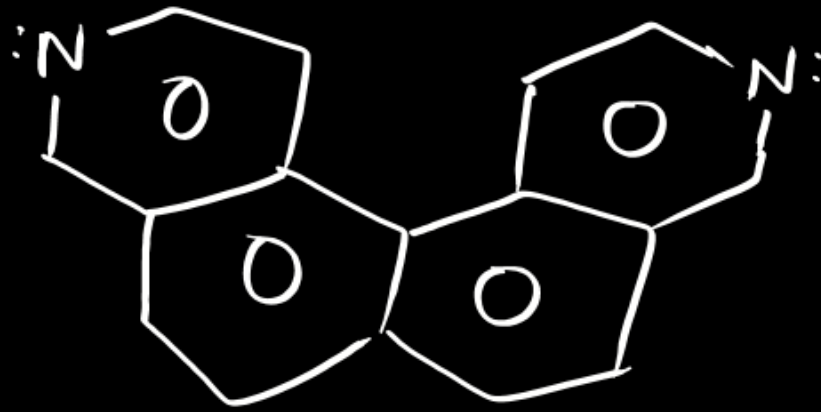
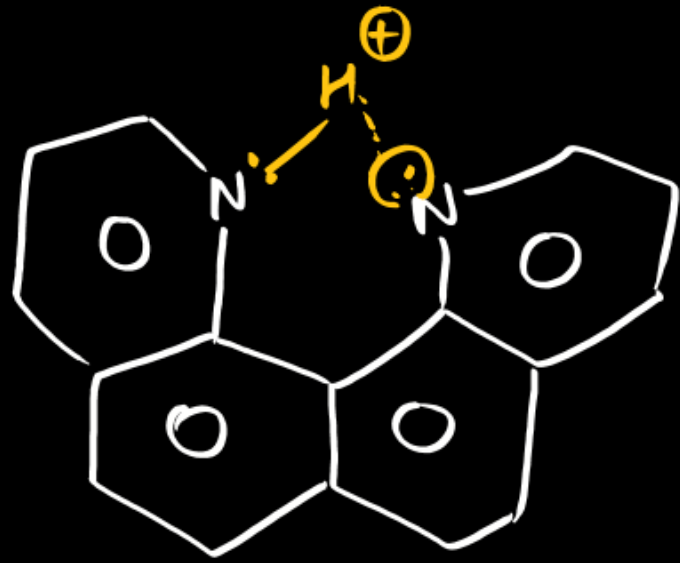
Ortho effect



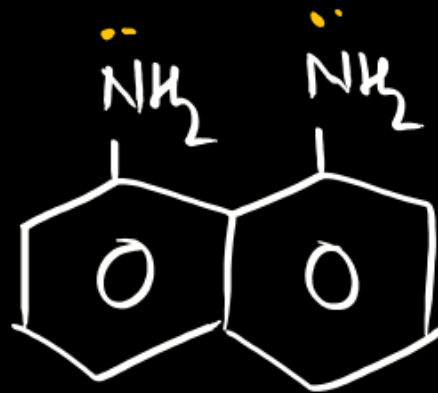
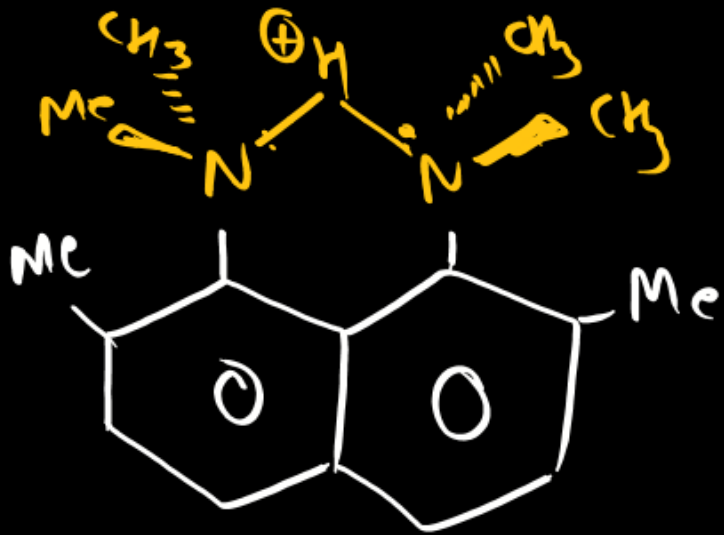
CATS (COMPETISHUN ADVANCED TRAINING SESSION)

Q. Giver order of basic strength.

①

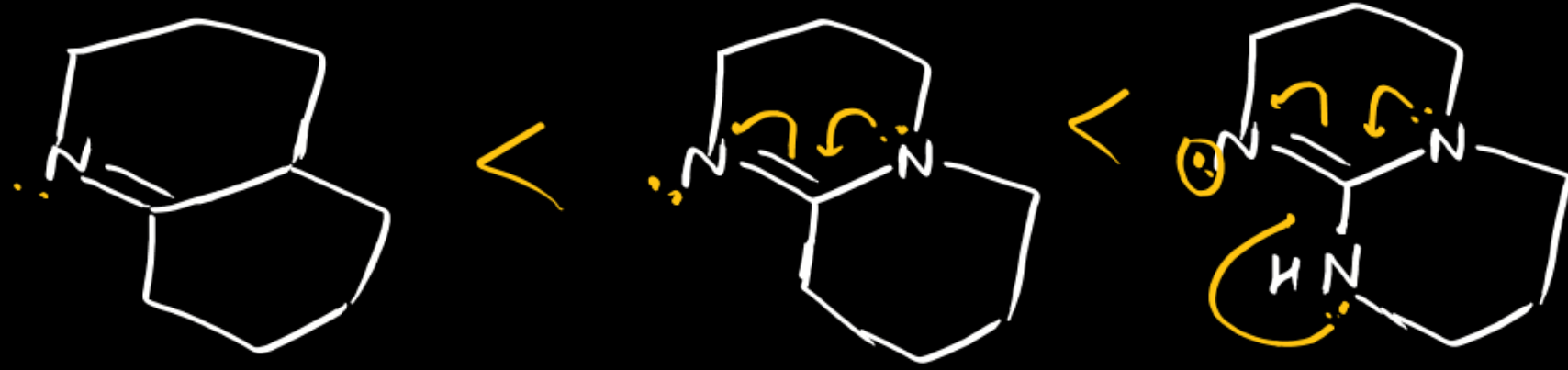


②

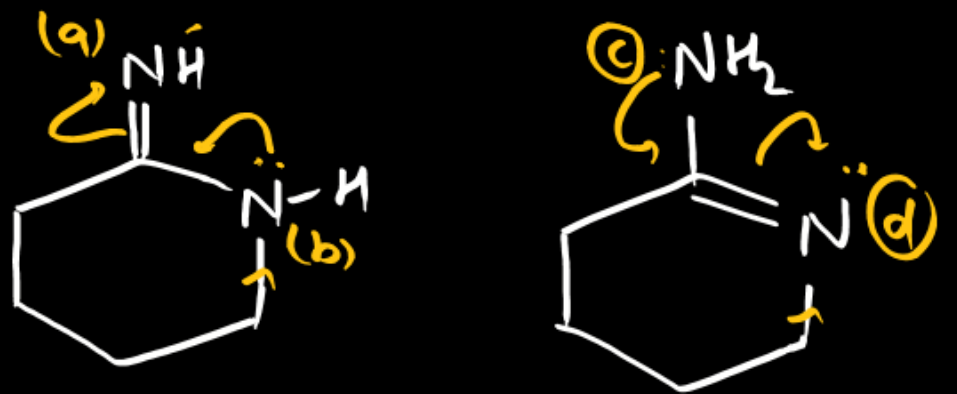


CATS (COMPETISHUN ADVANCED TRAINING SESSION)

③



(4)



$d > a > b > c$

(5)

